Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Engineering & Technology
B.Tech. | Sem-5 | Summer-2025

Course Code	:	2301CS501	**Date**	:	05-05-2025
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define Computer Network with example & List application used in real life.	**4**
	(B)	Compare LAN, MAN & WAN.	**3**
		OR	
		Explain star topology with example.	
	(C)	Draw OSI Layer with each layer functionality.	**7**
		OR	
		Compare OSI layer and TCP/IP model.	
Q.2	**(A)**	What is importance of application layer in computer network? Justify.	**4**
	(B)	State the port of below: of below: 1) DNS, 2) SMTP, 3) HTTPS.	**3**
		OR	
		Write a short note on FTP.	
	(C)	Discuss how email works using SMTP protocol.	**7**
		OR	
		Explain persistent HTTP with suitable example.	
Q.3	**(A)**	Explain checksum with any example.	**4**
	(B)	Draw UDP header.	**3**
		OR	
		Draw TCP header.	
	(C)	Explain rdt 2.1 with diagram.	**7**
		OR	
		Explain rdt 3.0 with diagram.	

Q.4	**(A)**	Define subnetting and find total subnet of IP address 192.168.0.1/26.	**4**
	(B)	Find the total number of host IP address 192.168.0.1/26.	**3**
		OR	
		Compare datagram network and virtual circuit network.	
	(C)	Draw IPv4 datagram format.	**7**
		OR	
		Draw IPv6 datagram format.	
Q.5	**(A)**	Explain parity check technique with any example.	**4**
	(B)	Write a short note on FDMA.	**3**
		OR	
		Write a short note on CDMA.	
	(C)	Explain CRC technique with any example.	**7**
		OR	
		Describe how ARP works with suitable example.	

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